货币政策与企业就业预期的刚性

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工作论文系列

本文研究欧洲央行货币政策意外如何影响企业的就业预期及随后的就业结果。我们将德国ifo企业调查的月度数据（2005年至2024年间每月覆盖约9000家企业）与高频货币政策意外相结合。通过比较在欧洲央行公告前与公告后不久接受调查的企业，我们隔离出政策冲击对预期的即时影响。

这与现有关于金融加速器的证据有所不同，后者侧重于已实现的就业或投资结果。这些模式表明，政策传导取决于企业特定的（财务）状况和劳动力市场法规。

因此，企业的就业预期为货币政策如何传导至劳动力市场提供了早期且直接的观察视角。然而，尽管关于货币政策对产出和通胀预期影响的文献日益增多（Bottone and Rosolia, 2019; Enders et al., 2019; Di Pace et al., 2025），但关于央行公告如何塑造企业就业预期、以及这些预期如何转化为实际就业的证据仍然有限。这一空白在微观层面尤为突出，因为企业层面的信念形成最终驱动着总体劳动力市场动态。

因此，我们能够评估中央银行是否只能通过需求变化间接影响就业，还是也能通过预期变化直接影响就业。

(Bahaj et al., 2022; Singh et al., 2023)，我们的结果表明，这种差异化反应源于企业如何形成其招聘计划，从而弥合了货币政策公告与实际就业变化之间的差距。此外，我们通过企业自我报告的财务困难和信贷获取条件直接衡量财务约束，而非依赖杠杆率或企业规模等资产负债表代理指标。

困难或报告过度约束。最后，我们的结果强调了劳动力市场刚性在新凯恩斯主义模型中的重要性。生产预期与就业预期调整之间的差距表明，存在招聘和解雇成本，以及寻找新员工和辞退现有员工的难度。Lechthaler等人（2010）在其新凯恩斯主义模型中证明了劳动力调整成本的相关性，该模型用于从货币政策冲击中推导出持续性的失业效应，以及就业创造与就业破坏之间的负相关关系。

Enders et al. (2019)的研究表明，企业的价格预期和生产预期会对欧洲央行的公告作出反应。我们在三个方向上有所不同：从产品市场转向劳动力市场，其中制度特征产生了根本不同的动态变化；从即时影响转向通过局部投影法揭示的完整动态路径，显示出就业预期的持续性远超生产预期；以及通过财务约束和劳动力市场约束来分析货币政策对企业预期的传导。Bottone和Rosolia（2019）、Ferrando等人（2022）以及Di Pace等人（2025）分别提供了货币政策对通胀预期、信贷获取预期和价格预期影响的补充证据。

(2020)报告了非常规政策效果不一的结果。Francis等人(2012)展示了美国各城市间政策的异质性，与之相关的是，de Groot等人(2023)记录了欧元区的一个空间维度：货币紧缩导致较贫困地区就业和劳动生产率持续下降，这表明劳动力市场调整中存在滞后效应，这与我们在企业招聘意向中记录到的持续性相呼应。此外，经济的部门构成影响货币政策向产出的传导强度(Hauptmeier等人，2025)。关于分配后果，Coglianese等人(2025)表明，货币紧缩提高了瑞典的失业率和劳动力市场不平等——这与我们发现裁员边际最终占据主导的结果相呼应。Groiss(2023)强调了就业创造渠道在塑造德国工资不平等中的作用，我们的招聘边际结果在企业层面证实了这一机制。与我们最接近的使用企业层面数据的研究是Bahaj等人(2022)和Singh等人(2023)，他们分别展示了企业特征如何塑造英国和美国货币政策对就业的影响。我们通过将研究从实际就业转向就业预期，

区分招聘和解雇意向，覆盖所有主要部门，并识别财务约束、信贷获取、工会覆盖率和最低工资在欧洲央行货币政策传导中的具体作用，来扩展这项工作。

数据中心（EBDC）。该数据自1980年起以月度频率覆盖制造业，其他部门的数据随后逐步可得。[3]自2005年起，所有部门均已纳入，因此我们将样本期设定为2005年至2024年。该调查覆盖面广泛，每月约收到9,000份回复，并在部门内具有代表性（Sauer et al., 2023）。[4]另一个优势在于调查的月度面板结构。我们不仅能够重复观察到同一批企业，而且对于一项自愿性调查而言，其回复率非常高，流失率则较低。[3]行业（包括制造业（IBS-IND, 2024）和建筑业（IBS-CON, 2024））、服务业（包括贸易业（IBS-TRA, 2024）和其他服务业（IBS-SERV, 2024））。

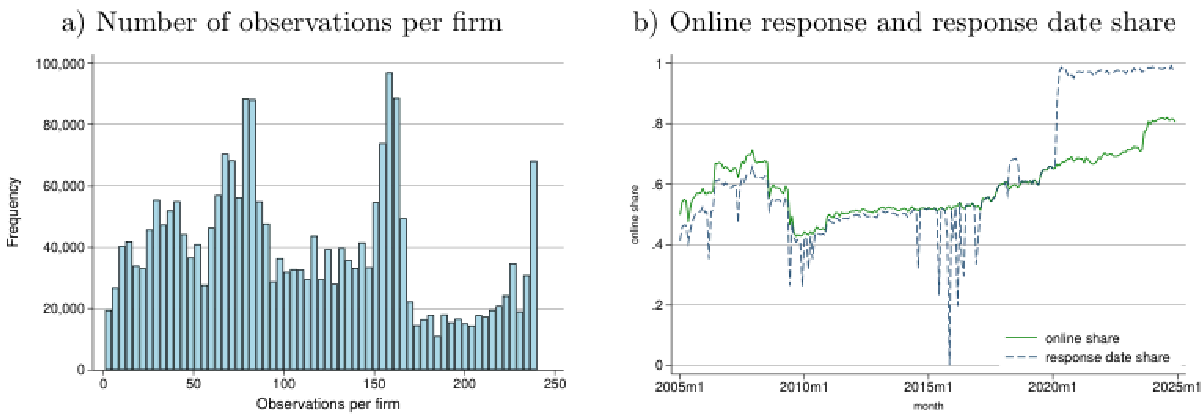

图1：ifo 调查——样本特征

a) 每家企业观察值数量 b) 在线回复及回复日期占比

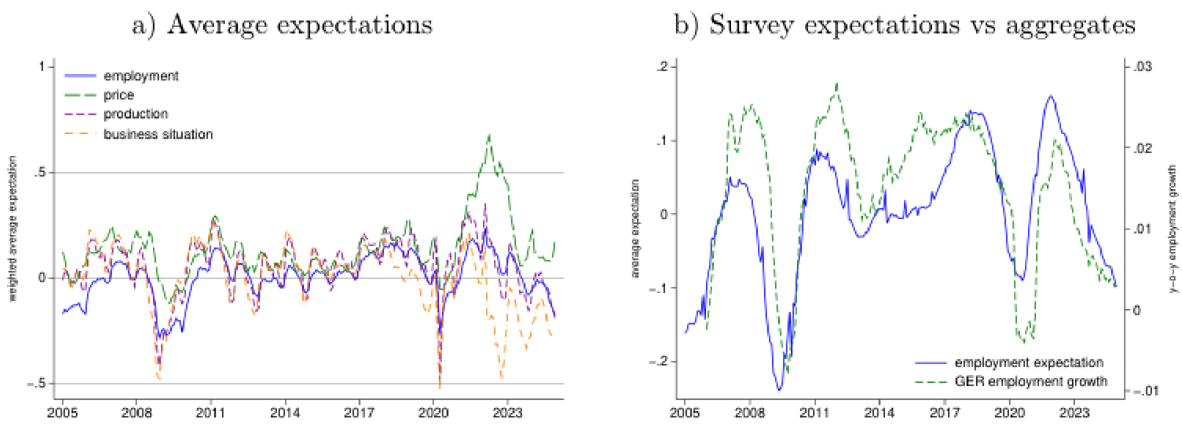

图2：就业预期

a) 平均预期 b) 调查预期与总量指标对比

因此，这允许对冲击进行外生性解释。我们通过检查政策冲击的自相关性以及其他冲击是否以显著且有意义的方式影响货币政策意外来检验外生性（Bauer and Swanson, 2023），而欧元区的情况并非如此（见表B.1）。为应对剩余担忧，我们在下一节估计即时预期调整时，利用了ifo调查的准随机响应日。

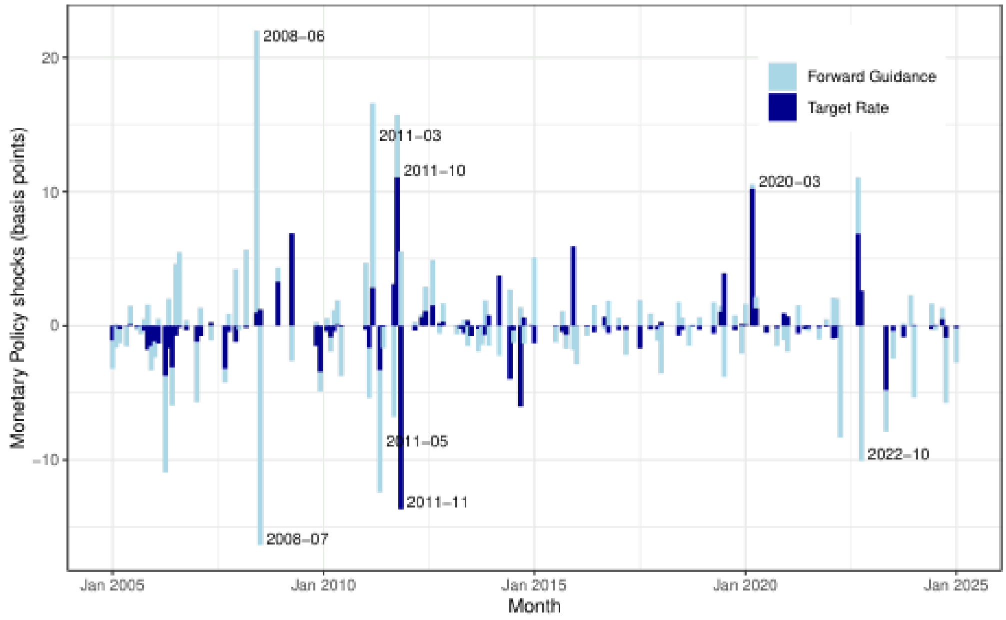

图3：高频货币政策冲击

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我们假设企业做出回应的时间与公告本身无关，这一假设得到了处理组和未处理组企业预期分布相似性的支持（见表A.2）。

表1展示了我们对企业货币政策意外即时预期响应的基准结果。第(1)列显示，目标利率意外上升25个基点会导致就业预期出现统计上显著的下降，幅度为0.13。具体而言，在货币紧缩之后，降低就业预期的企业比增加就业预期的企业多出13%。

就业预期的均值为0.02，标准差为0.54。系数范围介于-2和+2之间。在极端情况下，若所有企业在25个基点政策冲击后计划增加劳动力（1）并转而减少劳动力（-1），则特定事件的系数将为-2。

Table 1: Immediate expectation responses

	employment expectation			production expectation		
	total	increase	decrease	total	increase	decrease
	(1)	(2)	(3)	(4)	(5)	(6)
TR shock × post	-0.1280***	-0.0943***	0.0296***	-0.2903***	-0.0836***	0.2046***
	(0.0120)	(0.0082)	(0.0083)	(0.0162)	(0.0099)	(0.0112)
Post dummy	0.0003	-0.0014*	-0.0014**	-0.0138***	-0.0044***	0.0096***
	(0.0010)	(0.0007)	(0.0007)	(0.0013)	(0.0009)	(0.0008)
Exp employment (t-1)	0.4766***	0.4118***	0.4858***			
	(0.0012)	(0.0016)	(0.0016)			
Exp production (t-1)				0.3468***	0.3140***	0.3320***
				(0.0013)	(0.0013)	(0.0015)
Exp business situation (t-1)	0.0576***	0.0312***	-0.0320***	0.1520***	0.0874***	-0.0752***
	(0.0008)	(0.0005)	(0.0006)	(0.0012)	(0.0008)	(0.0007)
Constant	0.0067***	0.0852***	0.0675***	0.0520***	0.1612***	0.1038***
	(0.0006)	(0.0005)	(0.0004)	(0.0008)	(0.0006)	(0.0005)
Firm FE	yes	yes	yes	yes	yes	yes
Meeting FE	yes	yes	yes	yes	yes	yes
Observations	209,399	209,399	209,399	209,799	209,799	209,799
R^2	0.5198	0.4800	0.4769	0.4481	0.4149	0.3910
Number of Meetings	200	200	200	200	200	200

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on employment and production expectations. Heteroskedasticity robust standard errors in the parenthesis. TR=Target Rate.
* p<0.10, ** p<0.05, *** p<0.001.

转向生产预期，第（4）列显示出一个显著更大的即时响应。在基准设定中，25个基点的目标利率意外会使生产预期下降0.29，而在加入额外控制变量后（见表D.3），这一降幅甚至更大。由于这两个预期序列均为有序变量，系数衡量的是调整计划的企业净占比，而非调整的绝对规模，因此最好通过调整可能性来比较。基于此，差距十分明显：在货币政策公告后，调整生产预期的企业数量几乎是调整就业预期的两倍。这一模式与劳动力市场比产品市场具有更大刚性的观点相符。这些刚性可能包括与招聘和解雇相关的调整成本，这与就业保护立法的严格程度以及企业特定人力资本的形成有关。这一差异并非源于两个变量在不同尺度上衡量。它们在企业间的均值和标准差非常相似，因此对两者进行标准化后，差距基本不变：25个基点的意外使标准化生产预期下降0.48，而就业预期下降0.25（见表D.1第（4）和（5）列），比率再次接近二比一。

第（5）列和第（6）列中三分法变量的分解表明，生产预期的下降主要源于负面预期的增加。与就业预期不同，企业在面对紧缩性货币政策冲击时，其主要的即时策略并非缩减扩张计划，而是减少现有生产。

企业预期的动态响应

这种持续性符合劳动力市场摩擦的特征——包括通知期、就业保护以及重组劳动力所需的时间——这些摩擦阻碍了企业像调整生产那样迅速地调整就业。

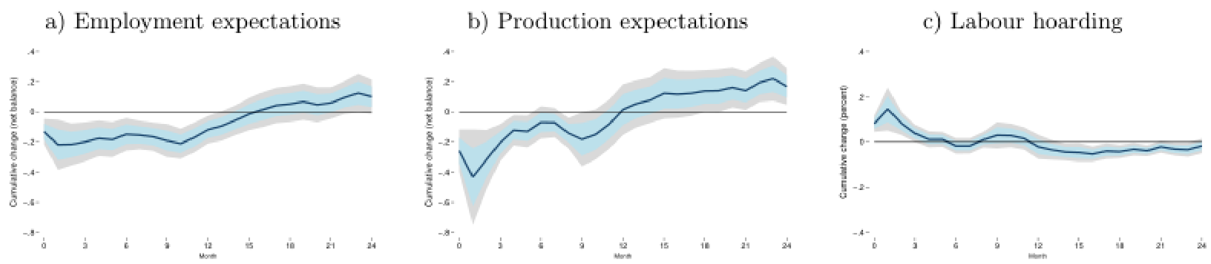

图4：动态预期响应

a) 就业预期

这表明货币政策在产品市场和劳动力市场中通过不同的渠道发挥作用，其中劳动力市场表现出明显更大的惯性。德国的劳动力市场制度，包括法定就业保护、最低工资以及企业职工委员会的普遍存在，可能解释了这种差异性的持续性。此外，自全球金融危机以来，面对持续的技术工人短缺，企业不愿裁员（Groiss and Sondermann, 2025），这可能进一步减缓了就业预期调整的速度。

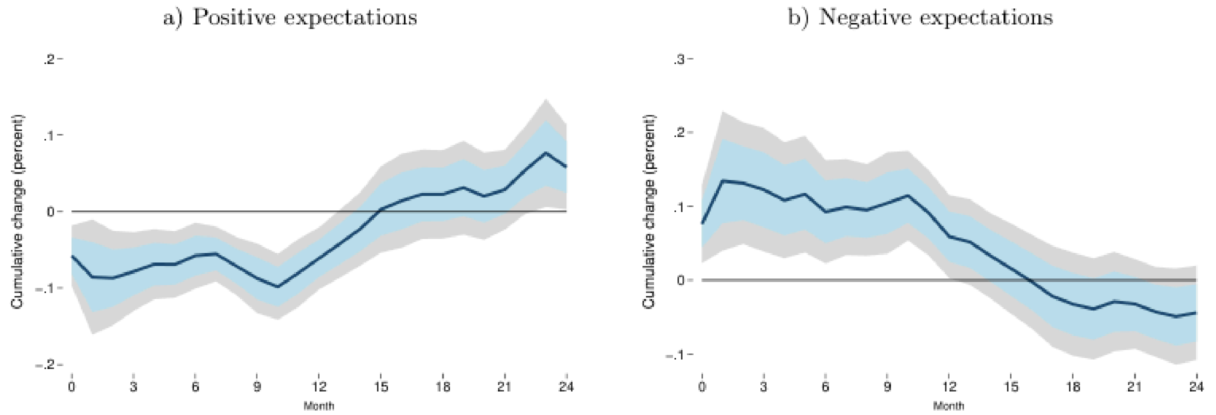

图5：招聘减少与裁员增加

a) 正面预期

Table 2: Employment growth response

	year-ahead employment growth			
	(1)	(2)	(3)	(4)
TR shock	-1.4200***	-0.5973***	-0.2636	-1.7360***
	(0.4055)	(0.0821)	(0.2405)	(0.3649)
Declining employment expectations (t-1)	-0.6123***	-0.3672***		
	(0.2237)	(0.0240)		
Increasing employment expectations (t-1)	0.3847***	0.2135***		
	(0.1480)	(0.0249)		
Employment growth (t-1)	0.8932***	0.9541***	0.8888***	0.9196***
	(0.0289)	(0.0028)	(0.0086)	(0.0061)
Employment growth (t-2)	0.0196	-0.0083***	0.0108	-0.0078
	(0.0248)	(0.0028)	(0.0081)	(0.0070)
Employment growth (t-3)	-0.0615***	-0.0568***	-0.0485***	-0.0361***
	(0.0117)	(0.0019)	(0.0082)	(0.0042)
Lagged policy shocks	yes	yes	yes	yes
Firm and calendar month FE	yes	yes	yes	yes
Observations	1,199,955	1,109,829	124,973	113,457
R^2	0.8169	0.8538	0.8391	0.8682
Number of Meetings	200	200	200	200

Notes: OLS regressions showing the monetary policy effect on 12-month employment growth. Firm cluster robust standard errors in the parenthesis. TR=Target Rate. * p<0.10, ** p<0.05, *** p<0.001.

我们提供一系列估计值而非单一估计值，因为这些估计应谨慎解读。企业层面的就业变量大多按年度频率报告，并由ifo研究所跨月转发，这引入了测量噪声。为捕捉企业观测值内由此引发的序列相关性，我们控制了

滞后就业增长，并估计企业聚类稳健标准误。第(1)列报告了使用与局部投影法相同样本和规格的估计结果，其中包括三期就业增长滞后项、三期货币政策冲击滞后项以及固定效应。鉴于净余额下降0.21，至少21%的企业将不得不下调其预期（如果没有企业上调预期的话）。假设这些企业每家计划裁员一名员工（见A.4），且德国约有340万家企业，这将导致71万个工作岗位的流失。与2024年德国员工总数（4200万）相比，这相当于就业减少1.7%，与图6中的总体就业响应相似。

预期的作用对货币政策传导同样重要。在第（3）列和第（4）列中，我们划分了样本，分别评估目标利率冲击对政策意外前就业预期为正（第（3）列）和就业预期为负（第（4）列）的企业实际就业增长的影响。对于就业预期为正的企业，紧缩性货币政策冲击的影响较小且统计上不显著。相比之下，对劳动力规模持悲观态度的企业则大幅削减员工，从而推动了第（2）列中整体样本的负系数。

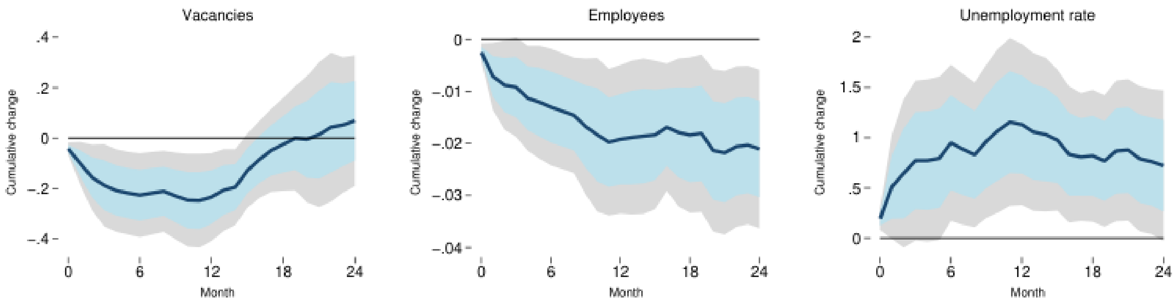

图6：总体劳动力市场响应

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正如Enders等人（2019）所述，信息效应可能随着冲击规模的增大而增强，从而逆转观测到的效应。因此，控制这些效应对于理解实际货币政策效果至关重要。17

Table 3: Employment expectation responses - nonlinearity and information effects

	employment expectation				
	(1)	(2)	(3)	(4)	(5)
TR shock × post	-0.0433*				
	(0.0256)				
Cubic TR shock × post	-0.6554***				
	(0.1729)				
JK monetary shock × post		-0.1207***			
		(0.0271)			
JK info shock × post		0.3343***			
		(0.0338)			
Positive TR shock × post			-0.1537***		
			(0.0132)		
Negative TR shock × post			0.0665		
			(0.0423)		
Policy rate change × post				0.1357	
				(0.1439)	
FG shock × post					0.0120
					(0.0085)
Firm FE	yes	yes	yes	yes	yes
Meeting FE	yes	yes	yes	yes	yes
Observations	209,399	193,480	209,399	209,399	209,399
R^2	0.5198	0.4670	0.5198	0.5197	0.5197
Number of Meetings	200	188	200	200	200

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on employment expectations. Heteroskedasticity robust standard errors in the parenthesis. (1) cubic effects, (2) Jarociński and Karadi (2020) policy surprises, (3) asymmetric effects, (4) policy rate change, (5) forward guidance shocks. TR=Target Rate, FG=Forward Guidance, JK=Jarocinski/Karadi. * p<0.10, ** p<0.05, *** p<0.001.

Di Pace等人（2025）比较了不同的货币政策衡量指标，以检验哪个指标对英国企业具有重要影响。他们发现，企业的价格预期会对实际政策利率变动和叙事冲击作出反应，但不会对金融市场中产生的“复杂”高频冲击作出反应。因此，我们在第（4）列中针对欧洲央行主要再融资操作利率的变动进行了回归分析。与Di Pace等人（2025）的发现不同，我们没有发现对企业预期的统计显著影响。政策利率变动的预期以及政策公告中所包含的信息效应似乎在德国企业中占据主导地位。然而，我们发现了与高频货币政策冲击显著且符合理论预期的响应，这与分析已实现企业层面就业的其他研究（如Bahaj等人，2022）一致，表明即使是小企业也会对金融市场变化作出反应。

第（5）列和第（6）列展示了多元逻辑回归的结果。鉴于因变量为具有三种可能结果的分类型变量，我们检验了研究发现在专门针对三分变量设计的回归方法下的稳健性。该方法将就业预期响应分解为正向和负向两个部分，与表1第（2）列和第（3）列的分解方式相对应。紧缩性货币政策冲击显著降低了就业预期上升的概率，同时显著提高了就业预期下降的概率。

Table 4: Employment expectation responses - Robustness

	employment expectation					
	(1)	(2)	(3)	(4)	(increase)	(decrease)
Target Rate shock × post	-0.1570***	-0.1281***	-0.1166***	-0.1003***	-0.7654***	0.7231***
	(0.0222)	(0.0122)	(0.0141)	(0.0303)	(0.1160)	(0.1112)
Further controls	no	no	yes	no	no	no
Firm FE	yes	yes	yes	yes	yes	yes
Meeting FE	yes	yes	yes	yes	yes	yes
Observations	341,792	209,399	117,371	209,399	209,399	209,399
R^2	0.4380	0.5198	0.5682	0.4655		
Number of Meetings	200	200	200	200	200	200

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on employment expectations. Heteroskedasticity robust standard errors in the parenthesis. Further controls include five additional lags in the respective expectation variable, current state of the business and number of employees (in log). (1) +/- 4 working day window, (2) set largest +/- shock to zero, (3) extra controls, (4) not weighted by firm-size, (5) multinomial logit: increasing workforce (vs unchanged), (6) multinomial logit: decreasing workforce (vs unchanged). TR=Target Rate. * p<0.10, ** p<0.05, *** p<0.001.

就业预期的下降与OLS估计结果一致。货币政策对生产预期的影响同样稳健，详见表D.3。

我们决定采用即时响应设定而非局部投影法框架，原因有二。首先，通过使用三重交互项，我们可以直接检验效应差异的统计显著性，而非比较脉冲响应函数，并控制时间固定效应。其次，更为重要的是，这保证了我们异质性维度的外生性。例如，企业规模可能在货币政策冲击后随时间发生变化，但在政策公告后的两天内不太可能调整。

相比之下，面临受限（即限制性）信贷获取的企业表现出显著为负的预期响应（-0.242）。这表明我们基线估计中的总体效应主要由面临财务约束的企业驱动，证实了货币政策的效应主要通过金融市场及其潜在约束进行传导。该设定中较小的样本量（截至2017年的136次会议）反映了调查中信贷获取问题的有限可得性。18

Table 5: Employment expectations - Financial and Labour Market rigidities

	employment expectation				
	(1)	(2)	(3)	(4)	(5)
TR shock × post	-0.1303***	-0.0050	-0.1269***	-0.4239***	0.0973*
	(0.0125)	(0.0208)	(0.0398)	(0.0281)	(0.0636)
TR × post × Financial difficulties (t-1)	-0.1367**				
	(0.0549)				
TR × post × Good credit access (t-3)		0.0389			
		(0.0465)			
TR × post × Reserved credit access (t-3)		-0.2422***			
		(0.0347)			
TR × post × $50-249$ empl			0.0153		
			(0.0532)		
TR × post × $250-999$ empl			0.0286		
			(0.0674)		
TR × post × >1000 empl			0.0073		
			(0.1084)		
TR × post × sectoral CBC share				0.0110***	
				(0.0010)	
TR × post × Minimum Wage					-0.1804***
					(0.0325)
Baseline controls	yes	yes	yes	yes	yes
Firm & Meeting FE	yes	yes	yes	yes	yes
Observations	201,319	78,221	180,516	166,865	26,788
R^2	0.5197	0.5181	0.4702	0.5195	0.6333
Number of Meetings	200	136	200	140	28

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on employment expectations. Heteroskedasticity robust standard errors in the parenthesis. Baseline category: (1) no financial difficulties, (2) normal credit access, (3) small firms (< 50 employees), (4) no collective bargaining coverage, (5) not affected by minimum wage, TR=Target Rate, CBC=collective bargaining coverage. * p<0.10, ** p<0.05, *** p<0.001.

劳动力市场刚性

对于规模差异，我们仅按部门规模而非企业规模对回归进行加权。将货币政策冲击与企业年龄交互，同样得到系数较小且统计上不显著的估计结果。

Table 6: Employment expectation - Constraints, labour hoarding and sectors

	employment expectation				
	(1)	(2)	(3)	(4)	(5)
TR shock × post	-0.0466***	-0.0431*	-0.0986***	-0.1438***	-0.3518***
	(0.0140)	(0.0254)	(0.0153)	(0.0126)	(0.1423)
TR × post × Services & Trade	-0.2020***				
	(0.0209)				
TR × post × Sectoral labour intensity		-0.0058***			
		(0.0015)			
TR × post × Constraints			-0.0510***		
			(0.0188)		
TR × post × Hoarding (t-1)				0.1392***	
				(0.0301)	
TR × post × Bus. state uncertainty (t-1)					0.0012
					(0.0019)
Baseline controls	yes	yes	yes	yes	yes
Firm & Meeting FE	yes	yes	yes	yes	yes
Observations	209,399	209,399	204,875	209,399	52,973
R^2	0.5198	0.5196	0.5197	0.5201	0.4425
Number of Meetings	200	200	200	200	60

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on employment expectations. Heteroskedasticity robust standard errors in the parenthesis. Baseline category: (1) industry (2) low labour intensity, (3) no constraints, (4) no hoarding, (5) low uncertainty, TR=Target Rate. * p<0.10, ** p<0.05, *** p<0.001.

第（3）列更广泛地探讨了经营约束的作用。对于不受约束的企业，基准效应为-0.099，而受约束的企业则表现出显著放大的响应。这表明，那些已在紧张条件下运营的企业——无论是由于劳动力市场法规、供给侧瓶颈、产能限制还是其他障碍——在制定就业计划时，对货币政策信号更为敏感。这一发现与理性疏忽模型一致，在该模型中，当调整的可能性受到限制且忽视冲击代价高昂时，主体会最强烈地更新其信念（Coibion et al., 2018）。同时，值得注意的是，受约束企业的系数并非像财务约束那样高度为负。这表明，在所有约束中，财务限制在货币政策冲击情况下对就业的影响最大。

结论

从政策角度看，就业预期响应的持续性表明，货币紧缩对劳动力市场的影响虽显现较慢，但持续时间远长于对产出的影响。由于企业在这些影响体现在总体统计数据之前几乎立即调整招聘计划，监测企业的就业预期可为货币政策是否如常传导至劳动力市场提供及时信号。

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Industrial productionIndustrial production index, Germany, calendar and seasonally adjustedBundesbank

Labour intensitySectoral ratio between employees and value added, 2-digit NACEEurostat

VariableDescriptionSource

- available since 2010IAB

Current business situationThe current state of business is: good, satisfiable, bad (1, 0, -1)IBS

ConstraintThe production is currently constrained (1, 0)IBS

Firm sizeFirm size category (1 = ¡50, 2 = 50-249, 3 = 250-999, 4¿1000)IBS

SectorEconomic sectors (# 79) - EBDC sector classificationIBS

monthsIBS

2018IBS

falling (1, 0, -1)IBS

months (1, 0, -1)IBS

-1)IBS

2005-2017IBS

are unchanged or increasing (1, 0)IBS

Employment expectationEmployment expectations for the next 3 months: increasing, stable, falling (1, 0, -1)IBS

wage (1, 0) - asked 2015/16 and 2022IBS

struction), services (incl. trade or other services)IBS

100 (high) - available since 2017IBS

Financial difficultyThe production is currently constrained by difficulties of financing (1, 0)IBS

adjustedifo

Credit accessCurrent willingness of banks to grant loans: accommodating, normal, reserved (1, 0, -1) - available

Minimum wageThe firm is (expected to be) affected by the introduction or increase in the statutory minimum

Business expectationThe business situation for the next 6 months will be: favourable, not changing, unfavourable (1, 0,

BS uncertaintyUncertainty with respect to commercial operations in the next 6 months on a scale from 0 (low) to

Production expectationProduction (construction, orders, demand) expectations for the next 3 months: increasing, stable,

Past employment changeEmployment in the previous month has increased, not changed, decreased (1, 0, -1) - available since

SurveyIndication in which IBS survey (sector) the firm takes part: industry (incl. manufacturing, con-

CBC shareSectoral collective bargaining agreement share, wage coverage related to trade unions negotiations

Price expectationPrices of own product/services are expected to increase, be unchanged, fall, during the next 3

HoardingIndicator for labour hoarding if production expectations are negative and employment expectations

ifo indexBusiness climate index, current business situation and expectations combined, all surveys, seasonally

EmploymentNumber of employees, reported once a year (October or November), forwarded to following 11

Table A.1: Variable description

Appendix

AData

Unemployment rateUnemployment registered in Germany, rate, calendar and seasonally adjustedBundesbank

VacanciesRegistered vacancies in Germany, calendar and seasonally adjustedBundesbank

MRO rateMain refinancing operations rate - ECB policy rateBundesbank

EmployeesNumber of employees subject to social security, calendar and seasonally adjustedBundesbank

CPIConsumer Price Index Germany, calendar and seasonally adjustedBundesbank

VariableDescriptionSource

Note: IBS = ifo Business Survey, covering the period January 2005 to December 2024 if not other stated, IAB = Establishment Panel

Table A.1: Variable description

A.1Variable harmonisation

While the ifo Business surveys for industry (incl. manufacturing, construction), and services (incl.trade and other services) are all provided by the ifo institute and have a lot in common, there arealso differences. Link (2020) provides a good overview about these differences and how to accountfor them. Here we want to give an overview about the questions (variables) used in our analysisand how they differ across sector (and time).

Employment expectation:

• Industry: (a) manufacturing: expectations for the next 3 months - The number of employeesassigned to the production of XY will be increasing, not changing, or decreasing. (01/2002-06/2018)Plans and Expectations for the next 3 months - The number of employees will be increasing,not changing, or decreasing. (since 07/2018)(b) construction: During the next 3-4 months the number of our employees will be increasing,not changing or decreasing.

• Services: (a) trade: plans and expectations for 3 months - Under elimination of seasonalfluctuations our number of employees will raise, not change, or fall. (03/2002-01/2007)Plans and expectations for the next 3 months - Under elimination of seasonal fluctuations ournumber of employees will raise, not change or fall. (02/2007-06/2018)Plans and expectations for the next 3 months - The number of our employees is expected toraise, not change, or fall. (07/2018-10/2018)The number of our employees is expected to raise, not change or fall. (since 11/2018)(b) other services: The number of employees will be increasing, not changing, or decreasingover the next 2-3 months. (10/2004-06/2018)Plans and expectations for the next 3 months - the number of our employees will presumablyraise, not change or fall. (since 07/2018)

Employment (annual):

• Industry: (a) manufacturing: number of employees - We occupy [. . .] persons in the wholecompany (domestic companies only).Thereof .....are assigned to the production of XY.(10/1999-02/2018)Number of employees - We employ [...] people throughout the company (domestic operationsonly). Of which the following are attributable to the product area. (since 02/2019)(b) construction: In year 20WW the average number of our employees altogether was....(12/2002-12/2017) In year 20WW the average number of our employees was...... (since 2018)

• Services: (a) trade: We occupy about .... persons in our company. (10/2000-02/2018)Number of persons employed - We employ in our company approx .... persons. (since 02/2019)(b) other services: Number of persons employed ....

Production expectation:

• Industry: (a) manufacturing: expectations for the next 3 months - Our domestic productionsof XY is expected to be increasing, not changing, decreasing or no production. (01/2002-06/2018)Plans and Expectations for the next 3 months - Our production is expected to be increasing,not changing, decreasing, or no production. (since 07/2018)(b) construction: During the next 3 months we will presumably build more, as much or lessthan/as in the last 3 months. (01/1991-06/2018)We will presumably build more, as much or less than in the past 3 months. (07/2018-10/2018)Our construction activities is presumably going to increase, stay roughly the same or decrease.(since 11/2018)

• Services: (a) trade: plans and expectations for 3 months - Compared to the same period oftime of previous year, our orders will likely be raising, not changing, or falling. (03/2002-01/2007)Plans and expectations for the next 3 months - Compared to the same period of time of pre-vious year, our orders will likely be raising, not changing, or falling. (02/2007-10/2018)Plans and expectations for the next 3 months - Our orders will likely be raising, not changing,falling. (since 11/2018)(b) other services: plans and expectations - the demand for our services/our turnover is ex-pected to increase, not change, decrease in the next 2-3 months. (10/2004-06/2018)Plans and expectations for the next 3 months - Our turnover will likely be raising, not chang-ing, falling. (since 07/2018)

Table A.2: Summary statics - expectations

Offline surveyOnline survey

DeclineUnchangedIncreaseDeclineUnchangedIncrease

Employment expectation0.120.750.120.120.740.13Past employment0.100.800.100.100.800.10Production expectation0.200.600.200.200.590.21Past production0.230.540.230.230.540.24Price expectation0.090.700.210.090.680.23Past prices0.110.750.140.090.750.16Credit access0.260.630.110.240.640.12

Untreated firmsTreated firms

DeclineUnchangedIncreaseDeclineUnchangedIncrease

Employment expectation0.110.750.140.120.750.13Past employment0.090.800.110.100.800.10Production expectation0.180.600.220.180.600.22Past production0.210.550.240.220.540.24Price expectation0.070.690.230.090.720.20Past prices0.080.760.160.100.770.14Credit access0.240.640.120.240.640.12

+/- 2 day windowout of window

DeclineUnchangedIncreaseDeclineUnchangedIncrease

Employment expectation0.120.750.140.130.750.12Past employment0.090.800.100.100.800.10Production expectation0.180.600.220.200.600.20Past production0.210.540.240.230.540.23Price expectation0.080.700.210.100.700.21Past prices0.090.760.150.110.750.14Credit access0.240.640.120.260.630.11

Note: Descriptive statistics based on the ifo Business Survey, comparing the expectations of firmsresponding online or offline (telephone, fax), the expectations of treated (Friday-Monday after apolicy announcement) versus untreated (Tuesday-Wednesday before a policy announcement) firms,and the expectations of firms within the +/- 2 working day window or out of the window.

Table A.3: Summary statics - size and sector

Untreated firmsTreated firms

ConstrServiceTradeManufConstrServiceTradeManuf

<50 employees11,96115,60212,75112,58711,65311,38815,05210,09950-249 employees11,0205,9194,19621,00611,2574,4155,27418,311250-999 employees1,8851,62197210,0721,9991,2871,4999,919>999 employees3477523964,2474375875384,511Total25,21323,89418,31547,91225,34617,67722,36342,840

Note: The table shows the number of firms in each survey to size-class cell and compares treated(Friday-Monday after a policy announcement) to untreated (Tuesday-Wednesday before a policyannouncement) firms.

Table A.4: Summary statics - hiring vs layoff (2023-2024)

meanSDperc 10perc 25medianperc 75perc 90N

Hiring next 3 months5.5876.190013733,017Hiring last 3 months6.3247.6200131027,697Layoffs next 3 months4.6141.770013532,975Layoffs last 3 months5.5440.840013827,660

Note: Descriptive statistics based on the ifo Business Survey, comparing the hiring and layoff expectationsof firms with the realised hiring and layoff in the last 3 months. Questions are only available quarterlysince 2023.

BIdentification

Figure B.1: ifo survey - responses

a) Responses per day of weekb) Responses per day of month

150000

industryconstructiontradeservices

300000

100000

200000

observations

observations

100000

50,000

0

MondayTuesday WednesdayThursdayFridaySaturdaySunday0

1 2 3 4 5 6 7 8 9 10111213141516171819202122232425262728293031

Notes: Descriptive statistics based on the ifo Business Survey. Panel a) shows the number of survey responses handed-in per day of the week. Panel b) shows the number of survey responses per day of the month, decomposed by thesurvey sector.

Table B.1: Exogeneity - monetary policy shocks

Dependent variable:

Target Rate ShockForward Guidance Shock

autocorrfinancialautocorrfinancial

Policy Shock, t-1-0.0805-0.3400***(0.0721)(0.0721)Policy Shock, t-2-0.0389-0.1475*(0.0723)(0.0767)Policy Shock, t-30.0329-0.0116(0.0723)(0.0765)Policy Shock, t-40.02540.0292(0.0721)(0.0723)ΔDAX-0.02600.0037(0.0164)(0.0245)ΔGerman yield curve-0.17360.4716(0.5176)(0.7742)ΔUS 3m gov bond yield0.1168-0.7963*(0.2689)(0.4022)ΔFX GER-US0.86652.7436(1.8222)(2.7254)ΔCISS1.1707-1.4006(0.8529)(1.2757)Constant-0.0376-0.2186-0.05060.2774(0.1439)(0.2403)(0.2048)(0.3595)

Observations200200200200R20.0090.0550.1070.033

Notes: Δ= t −1 to t −90, frequency = daily.∗p<0.1; ∗∗p<0.05; ∗∗∗p<0.01.

CAggregate dynamics

To show the consistency of our empirical framework with the existing literature and to link ourfirm-specific findings on employment expectations to aggregate labour market outcomes, we showaggregate dynamic responses to monetary policy changes. We are doing so by estimating impulseresponse functions (IRF) using a local projection (LP) framework (Jord`a, 2005). At the aggregatelevel we estimate:

s=6X

yt+h = αh + βhεp

t +

γhyt−s + ΓhZt−1 + ut,h = 0, ..., 24(4)

s=1

where yt+h is our labour market variable of interest at horizon h. εp

t captures our high-frequencymonetary policy shocks, either the Target Rate shock or the Forward Guidance shock. αh is theintercept and Zt−1 includes both lagged policy shocks (and further macro controls for robustnesschecks). We also include six lags of the dependent variable to control for the persistence in theseaggregate indicators. We calculate heteroskedasticity and autocorrelation robust Newey-West stan-dard errors to represent the uncertainty around our estimates.

The upper row of Figure C.1 shows the responses of the industry production index, the ifobusiness climate index and the consumer price index of Germany to a 25 basis point point increaseto the conventional Target Rate shock. In line with theory and the existing literature, we see asignificant decline in the real economy and a weak but negative effect on inflation (e.g. Jaroci´nski andKaradi, 2020). More related to our analysis are the aggregate monetary policy effects to Germanlabour market variables.While vacancies and, with a bit delay, employees decline following acontractionary monetary policy shocks, the unemployment rate increases up to one percentagepoint.

Figure C.1: Aggregate responses to monetary policy shocks

Industry production

ifo index

Consumer price index

.1

.1

.02

0

Cumulative change

Cumulative change

Cumulative change

0

0

−.1

−.1

−.02

−.2

−.2

−.04

−.3

−.3

−.06

06121824

06121824

06121824

Month

Month

Month

Vacancies

Employees

Unemployment rate

.4

2

0

Cumulative change (pp)

1.5

.2

Cumulative change

Cumulative change

−.01

1

0

−.02

.5

−.2

−.03

0

−.4

−.04

06121824

06121824

06121824

Month

Month

Month

Notes: This figure shows the impulse responses of aggregate macro indicators to a 25 basis point Target Rate shock.It depicts the relative cumulative growth in percent over the period 2005-2024 (change in unemployment rate inpercentage points). The shaded areas depict the 68% and 90% confidence bands, based on Newey-West standarderrors.

DMonetary Policy Effects - Robustness

Table D.1: Immediate expectation responses - Labour hoarding

labour hoardingemployment expproduction exp(production - employment gap)(standardised)(standardised)(1)(2)(3)(4)(5)TR shock × post0.1148∗∗∗0.1353∗∗∗0.1536∗∗∗-0.2512∗∗∗-0.4791∗∗∗

(0.0114)(0.0102)(0.0102)(0.0239)(0.0268)Post dummy0.0067∗∗∗0.0064∗∗∗0.0060∗∗∗0.0005-0.0228∗∗∗

(0.0009)(0.0006)(0.0007)(0.0020)(0.0022)Expectation employment (t-1)0.1434∗∗∗0.0514∗∗∗0.0568∗∗∗0.4766∗∗∗

(0.0008)(0.0006)(0.0006)(0.0012)Expectation production (t-1)0.3468∗∗∗

(0.0013)Expectation business situation (t-1)-0.0410∗∗∗-0.0363∗∗∗-0.0366∗∗∗0.1131∗∗∗0.2508∗∗∗

(0.0007)(0.0006)(0.0006)(0.0016)(0.0020)Constant0.1450∗∗∗0.0944∗∗∗0.0980∗∗∗0.0879∗∗∗0.1363∗∗∗

(0.0005)(0.0005)(0.0005)(0.0013)(0.0014)Firm FEyesyesyesyesyesMeeting FEyesyesyesyesyesObservations205,342205,342205,342209,399209,799R20.24160.24500.25030.51980.4481Number of Meetings200200200200200

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on labourhoarding indicators. (1) 1 if increasing/unchanged employment expectations and decreasing production ex-pectations or increasing employment expectations and unchanged production expectations, (2) 1 if unchangedemployment expectations and negative production expectations, (3) 1 if increasing/unchanged employmentexpectations and decreasing production expectations. (4) standardised employment expectations, (5) stan-dardised production expectations. Heteroskedasticity robust standard errors in the parenthesis. TR=TargetRate.* p<0.10, ** p<0.05, *** p<0.001

Table D.2: Production expectation responses - nonlinearity and information effects

production expectation(1)(2)(3)(4)(5)TR shock × post0.0189(0.0313)Cubic TR shock × post-2.4234∗∗∗

(0.2221)JK monetary shock × post-0.3072∗∗∗

(0.0348)JK info shock × post0.0835∗

(0.0451)Positive TR shock × post-0.3297∗∗∗

(0.0180)Negative TR shock × post-0.0236(0.0450)Policy rate change × post-0.3656∗∗

(0.1805)FG shock × post-0.0146(0.0108)Firm FEyesyesyesyesyesMeeting FEyesyesyesyesyesObservations209,799193,015209,799209,799209,799R20.44810.44530.44720.44790.4479Number of Meetings200188200200200

Notes:OLS regressions showing the monetary policy effect of treated (=post) vs non-treatedfirms on production expectations.Heteroskedasticity robust standard errors in the parenthe-sis. (1) cubic effects, (2) Jaroci´nski and Karadi (2020) policy surprises, (3) asymmetric effects,(4) policy rate change, (5) forward guidance shocks. TR=Target Rate, FG=Forward Guidance,JK=Jarocinski/Karadi.* p<0.10, ** p<0.05, *** p<0.001.

Table D.3: Production expectation responses - Robustness

production expectation(1)(2)(3)(4)(increase)(decrease)Target Rate shock × post-0.4041∗∗∗-0.2903∗∗∗-0.3176∗∗∗-0.1548∗∗∗-0.3905∗∗∗1.3921∗∗∗

(0.0301)(0.0162)(0.0492)(0.0383)(0.0937)(0.0875)

Further controlsnonoyesnononoFirm FEyesyesyesyesyesyesMeeting FEyesyesyesyesyesyesObservations312,484209,799119,297209,799209,799209,799R20.34700.44810.47560.3537Number of Meetings200200200200200200

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms onproduction expectations. Heteroskedasticity robust standard errors in the parenthesis. Further controlsinclude five additional lags in the respective expectation variable, current state of the business and numberof employees (in log). (1) +/- 4 working day window, (2) set largest +/- shock to zero, (3) extra controls, (4)not weighted by firm-size, (5) multinomial logit: increase (vs unchanged), (6) multinomial logit: decrease(vs unchanged). TR=Target Rate.* p<0.10, ** p<0.05, *** p<0.001.

Figure D.1: Dynamic expectation responses - Split-panel jackknife

a) employment expectationsb) production expectations

−.8−.6−.4−.20.2.4Cumulative change (net balance)

−.8−.6−.4−.20.2.4Cumulative change (net balance)

06121824Month

06121824Month

Notes: This figure shows the responses of firm-specific employment expectations (left) and production expectations(negative) to a 25 basis point Target Rate shock. The blue line depicts the OLS estimation, while the green linedepicts the split-panel jackknife estimator. The shaded areas depict the 68% and 90% confidence bands, based onfirm and time cluster robust standard errors.

Figure D.2: Dynamic expectation responses - Driscoll and Kraay (1998) SE

a) employment expectationsb) production expectations

−.8−.6−.4−.20.2.4Cumulative change (net balance)

.4

Cumulative change (net balance)

.2

0

−.2

−.4

−.6

−.8

03691215182124Month

03691215182124

Month

Notes: This figure shows the responses of firm-specific employment expectations (left) and production expectations(negative) to a 25 basis point Target Rate shock. The shaded areas depict the 68% and 90% confidence bands, basedon Driscoll and Kraay (1998) standard errors.

Figure D.3: Dynamic expectation responses - Jaroci´nski and Karadi (2020)

a) employment expectationsb) production expectations

.4

.4

.2

Cumulative change (net balance)

.2

Cumulative change

0

0

−.2

−.2

−.4

−.4

−.6

−.6

−.8

−.8

03691215182124

03691215182124

Month

Month

Notes: This figure shows the responses of firm-specific employment expectations (left) and production expectations(negative) to a 25 basis point pure monetary policy shock (Jaroci´nski and Karadi, 2020). The shaded areas depictthe 68% and 90% confidence bands, based on firm and time cluster robust standard errors.

Figure D.4: Dynamic expectation responses - Forwarding controls

a) Employment expectationsb) Production expectations

.4

.4

Cumulative change (net balance)

Cumulative change (net balance)

.2

.2

0

0

−.2

−.2

−.4

−.4

−.6

−.6

−.8

−.8

03691215182124

03691215182124

Month

Month

Notes: This figure shows the impulse responses of firm-specific employment expectation (left) and production expectations (right) to a 25 bp Target Rate shock. Additional (t+h-1) forwarded expectations are added to control thepersistence in the expectation variables. The shaded areas depict the 68% and 90% confidence bands, based on firmand time cluster robust standard errors.

Figure D.5: Dynamic expectation responses - production

a) Positive expectationsb) Negative expectations

.6

.2

Cumulative change (percent)

Cumulative change (percent)

.4

.1

0

.2

−.1

0

−.2

−.2

03691215182124

03691215182124

Month

Month

Notes: This figure shows the responses of firm-specific positive production expectations (left) and negative productionexpectations (negative) to a 25 basis point Target Rate shock. The shaded areas depict the 68% and 90% confidencebands, based on firm and time cluster robust standard errors.

Table D.4: Production expectations - Financial and Labour Market rigidities

production expectation(1)(2)(3)(4)(5)TR × post-0.3176∗∗∗-0.0054∗∗-0.1935∗∗∗-0.6604∗∗∗-0.3192∗∗∗

(0.0167)(0.0023)(0.0519)(0.0363)(0.0436)TR × post × Financial difficulties (t-1)-0.0163∗∗∗

(0.0059)TR × post × Good credit access (t-3)-0.1199∗∗

(0.0544)TR × post × Reserved credit access (t-3)-0.2055∗∗∗

(0.0471)TR × post × 50 –249 empl0.0455(0.0690)TR × post × 250 –999 empl0.0104(0.0909)TR × post × > 1000 empl0.0429(0.1329)TR × post × sectoral CBC share0.0126∗∗∗

(0.0013)TR × post × Minimum Wage-0.4326∗∗∗

(0.1393)Baseline controlsyesyesyesyesyesFirm & Meeting FEyesyesyesyesyesObservations201,70283,884180,863167,49926,927R20.44820.44600.44100.45230.5810Number of Meetings20013620014028

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on pro-duction expectations. Heteroskedasticity robust standard errors in the parenthesis. Baseline category: (1) nofinancial difficulties, (2) normal credit access, (3) small firms (< 50 employees), (4) no collective bargainingcoverage, (5) not affected by minimum wage, TR=Target Rate, CBC=collective bargaining coverage.*p<0.10, ** p<0.05, *** p<0.001.

Table D.5: Production expectation - Constraints, labour hoarding and sectors

production expectation(1)(2)(3)(4)(5)TR shock × post-0.2691∗∗∗-0.3231∗∗∗-0.3283∗∗∗-0.1716∗∗∗-0.2148∗∗∗

(0.0195)(0.0323)(0.0200)(0.0149)(0.0780)TR × post × Services & Trade-0.0065∗∗∗

(0.0022)TR × post × Sectoral labour intensity0.0017(0.0019)TR × post × Constraints-0.1069∗∗∗

(0.0198)TR × post × Hoarding (t-1)-0.1416∗∗∗

(0.0244)TR × post × Bus. state uncertainty (t-1)-0.0051∗∗∗

(0.0010)Baseline controlsyesyesyesyesyesFirm & Meeting FEyesyesyesyesyesObservations209,799198,027205,245207,09053,667R20.44810.44600.44830.58000.5158Number of Meetings20020020020060

Notes: OLS regressions showing the monetary policy effect of treated (=post) vs non-treated firms on pro-duction expectations. Heteroskedasticity robust standard errors in the parenthesis. Baseline category: (1)industry, (2) low labour intensity, (3) no constraints, (4) no hoarding, (5) low uncertainty, TR=Target Rate.* p<0.10, ** p<0.05, *** p<0.001.

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